

Stage 2 · Dense Per-Material Reconstruction

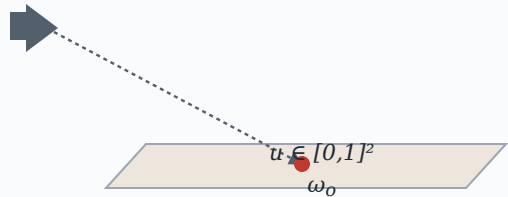
Target views → rays

view k = 1
HDR · ~400 views

view k = 2
HDR · ~400 views

view k = N
HDR · ~400 views

cast ray → UV on mesh



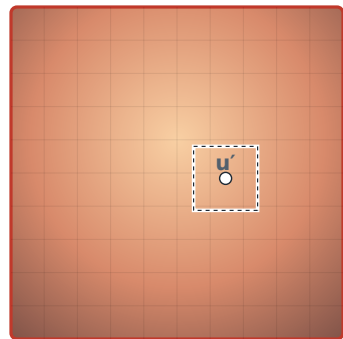
G_ϕ + Dense latent texture T_z

view-dependent displacement

G_ϕ
 $SH(\omega_o) \rightarrow \tanh$
small MLP

$$u' = u + \Delta u$$

dense latent texture T_z
2048 × 2048 × D



$$\text{progressive blur } \sigma(t) = \sigma_0 \cdot 2^{-t/T^{1/2}}$$

▲ TRAINABLE

Local frame & encoding

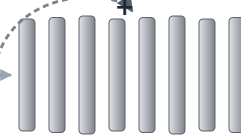
Gram-Schmidt

$$\begin{aligned}\tilde{n} &= \hat{n} \\ \tilde{t} &= (t - (t \cdot \tilde{n})\tilde{n})^\vee \\ \tilde{b} &= \tilde{n} \times \tilde{t}\end{aligned}$$

SH encoding



MLP decoder D_θ

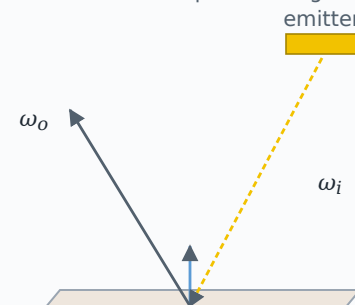


* FROZEN

Renderer → $\hat{I}$

forward renderer

area-emitter path tracing



→ $\hat{I}$ (predicted)